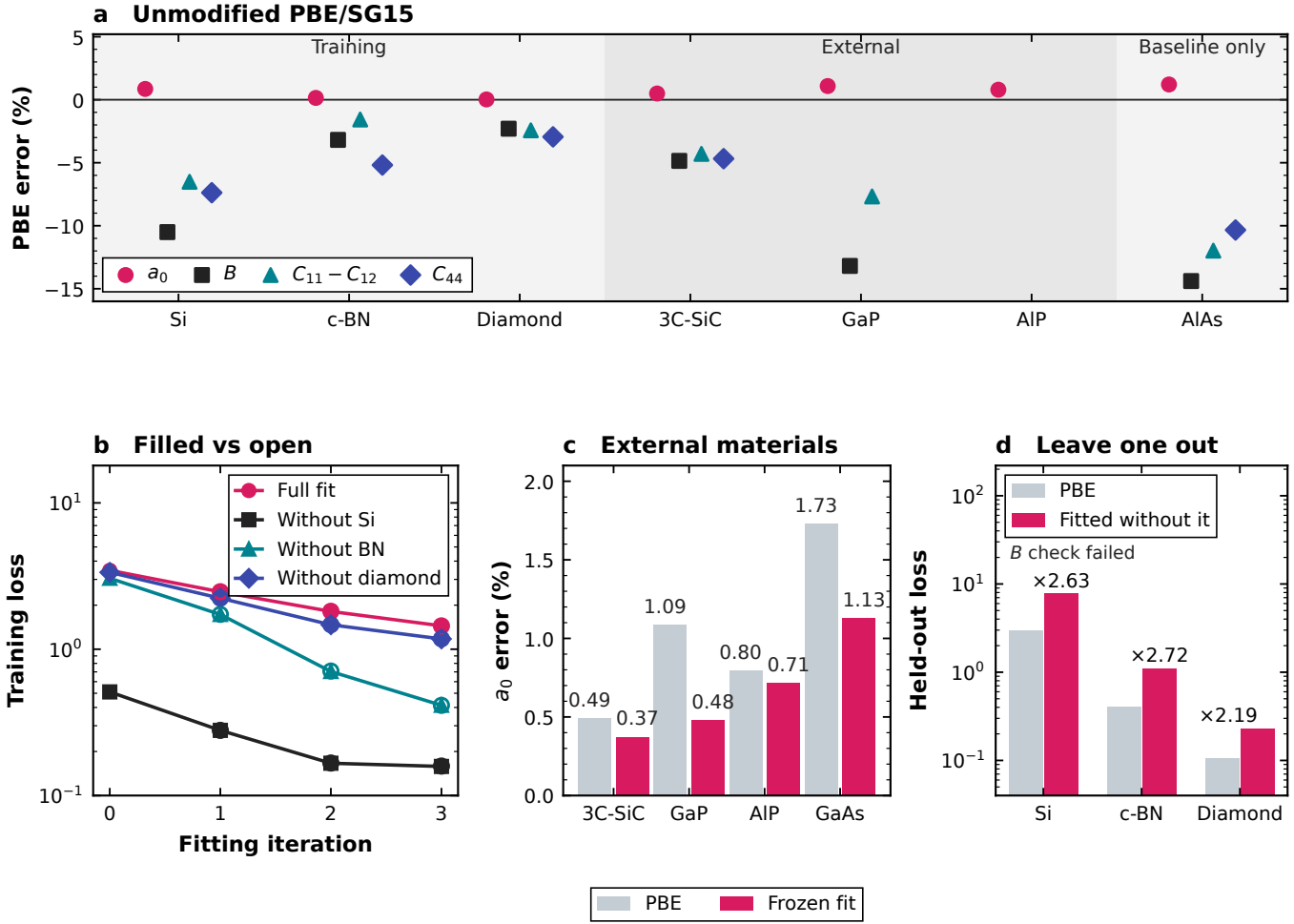
a Unmodified PBE/SG15
PBE error (%)
5
0
−5
−10
−15
Training
External
Baseline only
a_0
B
$C_{11}-C_{12}$
C_{44}
Si
c-BN
Diamond
3C-SiC
GaP
AlP
AlAs
b Filled vs open
Training loss
10^1
10^0
10^{-1}
Full fit
Without Si
Without BN
Without diamond
0
1
2
3
Fitting iteration
c External materials
a_0 error (%)
2.0
1.5
1.0
0.5
0.0
0.49
0.37
1.09
0.48
0.80
0.71
1.73
1.13
3C-SiC
GaP
AlP
GaAs
d Leave one out
Held-out loss
10^2
10^1
10^0
10^{-1}
PBE
Fitted without it
B check failed
×2.63
×2.72
×2.19
Si
c-BN
Diamond
PBE
Frozen fit